Banoth Arun

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Education

National Institute of Technology Patna **Expected June 2026**

B.Tech in Computer Science and Engineering | CGPA: 6.8/10.0

Technical Skills

LLMs and Generative AI: LangChain, LangGraph, Agentic Workflows, AI Agents, LLM APIs (OpenAI, Anthropic Claude, Gemini), Prompt Engineering, ReAct Prompting, Function Calling, JSON Outputs.

RAG and Vector Databases: RAG Pipelines, ChromaDB, Pinecone, Weaviate, FAISS, Text Embeddings, Semantic Search, Document Chunking, Retrieval-Augmented Generation.

Python and Frameworks: Python (OOP, DSA, Functions), FastAPI, Flask, PyTorch, Pandas, NumPy, Playwright, Automation Scripting, Jupyter Notebooks, Google Colab.

AI Dev Tools: Claude Code, GitHub Copilot, Cursor, Vibe Coding, AI-assisted Development, HuggingFace (familiar).

Tools and Technologies: MySQL, Node.js, MongoDB, React.js, Express.js, Visual Studio Code, Git, GitHub, Android Studio.

Soft Skills: Proactive Self-Learner, Curious Mindset, Initiative, Coachability, Team Collaboration, Communication, Adaptability.

Core Competencies: DSA, DBMS, OOPs, Neural Networks, Supervised Learning, Operating Systems, Agile, SDLC.

Software Development Projects

Multimodal Phishing Detection using GenAI (GitHub) **2025**

Tech Stack: *Python, PyTorch, RoBERTa (LLM), EasyOCR, Playwright, LangChain-style Pipeline*

- Built a **LangChain-style agentic AI pipeline** with **multi-step LLM reasoning, tool usage, and ReAct-style prompting** — self-learned RoBERTa, EasyOCR, and PyTorch from scratch, reflecting **proactive, curious, self-driven** AI engineering
- Implemented a **RAG-inspired memory and retrieval architecture** for dynamic feature storage across 34,000+ URLs; applied **FastAPI-style REST endpoints** and **JSON-structured LLM outputs** — achieving F1-score of 94.3% and 60% processing time reduction
- Wrote and iterated **prompts for LLM-based classification**; ran **model experiments**, documented results, and identified failure patterns — used **Git and GitHub** for version control with comprehensive **technical documentation**

Village Smart Shop Platform (Live Website - Active Village Users) **2026**

Tech Stack: *HTML, CSS, JavaScript, Firebase Firestore, Firebase Hosting*

- Proactively built and shipped a **real-world AI-ready full-stack application** from idea to production independently — exemplifying **self-motivated, initiative-driven** engineering with complete product lifecycle ownership
- Implemented **data ingestion, cleaning, and semantic-search-style retrieval** using Firebase Firestore — analogous to **RAG pipeline and vector database** architecture for live product, price, and stock management
- Wrote clear **technical documentation and utility scripts**; applied feedback rapidly — demonstrating strong **coachability, communication, and continuous improvement** mindset in a live production environment

Wanderlust - E-Commerce Property Platform (GitHub) **2025**

Tech Stack: *Python, Node.js, React.js, MongoDB, JavaScript, HTML5, CSS3, Tailwind CSS*

- Built **20+ REST API endpoints** with **FastAPI-compatible Python backend architecture**; designed MongoDB schema with optimized queries reducing response time by **40%** — backend fundamentals ready for **LangGraph and LangChain** agentic integration
- Wrote **Python automation and utility scripts** for data processing workflows; maintained **Git/GitHub** version control with structured branching and collaborative development — followed **Agile methodology** with code reviews and sprint planning
- Proactively stayed updated with **latest AI tools and LLM frameworks**; integrated modern front-end build tools and wrote **well-documented, production-ready code** with comprehensive technical documentation

Professional Experience

Machine Learning Intern – EI Systems **May 2025 – Jul 2025**

- Applied **Python, OOP, and ML/AI concepts** to build a production NLP pipeline — ran **LLM API-style model experiments**, performed **data collection, cleaning, and preprocessing**, documented results and identified failure patterns under senior guidance
- Explored **Prompt Engineering techniques** and **RAG-style information retrieval**; built **REST APIs** with optimized database queries reducing response time by 40% — used **Git/GitHub** for version control
- Participated in **Agile standups, sprint planning, and code reviews** — demonstrated **proactive learning, initiative, coachability, and positive attitude** toward all tasks assigned by senior engineers

Achievements and Certifications

- **Amazon MLSchool 2025:** Selected from 160,000+ applicants (1.8% acceptance rate) for advanced ML training covering **NLP, LLMs, Generative AI, computer vision, and production ML systems** — reflects genuine **curiosity and proactive self-learning** in AI
- **Amazon HackOn 2025:** Achieved Semifinalist position showcasing strong **AI problem-solving, agentic thinking, and fast-paced engineering** skills.
- **Flipkart Grid 7.0 2025:** Qualified as Semifinalist among top performers in a nationwide **software development and AI engineering** challenge.
- **HP Power Lab 2.0 – Semi-Finalist:** Cleared two competitive rounds showcasing analytical thinking and practical problem-solving. (Certificate)
- **HackerRank:** Cleared global skill certification with core Java fundamentals (Certificate)